



DEMO WEB SHOP AUTOMATION TESTING FRAMEWORK USING SELENIUM, CUCUMBER, TESTNG AND JAVA

CAPSTONE PROJECT REPORT

Submitted By

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ABSTRACT

Software testing plays a critical role in ensuring the quality, reliability, and functionality of web applications. Manual testing of complex e-commerce websites is time-consuming and prone to human error. To overcome these challenges, automation testing frameworks are widely used.

This project presents an Automation Testing Framework developed for the Demo Web Shop application using Selenium WebDriver, Java, Cucumber BDD, and TestNG. The framework adopts the Page Object Model (POM) design pattern to ensure maintainability, scalability, and reusability of test scripts. The framework automates important user workflows such as registration, login, product search, product comparison, wishlist management, shopping cart operations, guest checkout, registered user checkout, and return request processing.

Additionally, reporting tools such as Allure Report and Extent Report are integrated to provide detailed execution results and improve test analysis. The framework successfully demonstrates modern automation testing practices and significantly reduces manual testing effort.

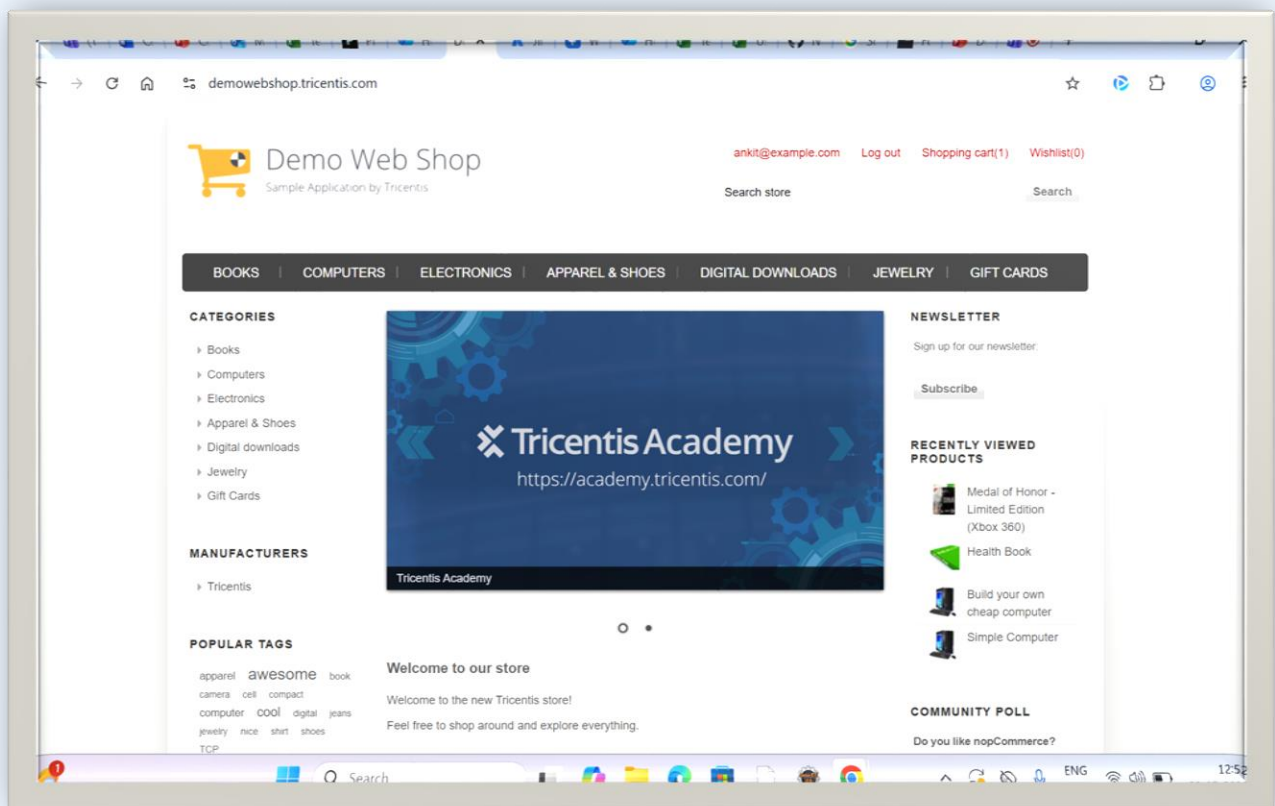


Figure 1. Demo Web Shop Application

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

The growth of e-commerce applications has increased the need for reliable and efficient testing solutions. Manual execution of repetitive test cases requires significant effort and often results in inconsistencies. Automation testing provides a solution by enabling quick, reliable, and repeatable test execution.

The Demo Web Shop Automation Framework is developed to automate major functionalities of an online shopping website. The framework uses Selenium WebDriver for browser automation, Java as the programming language, Cucumber for BDD implementation, and TestNG for execution management.

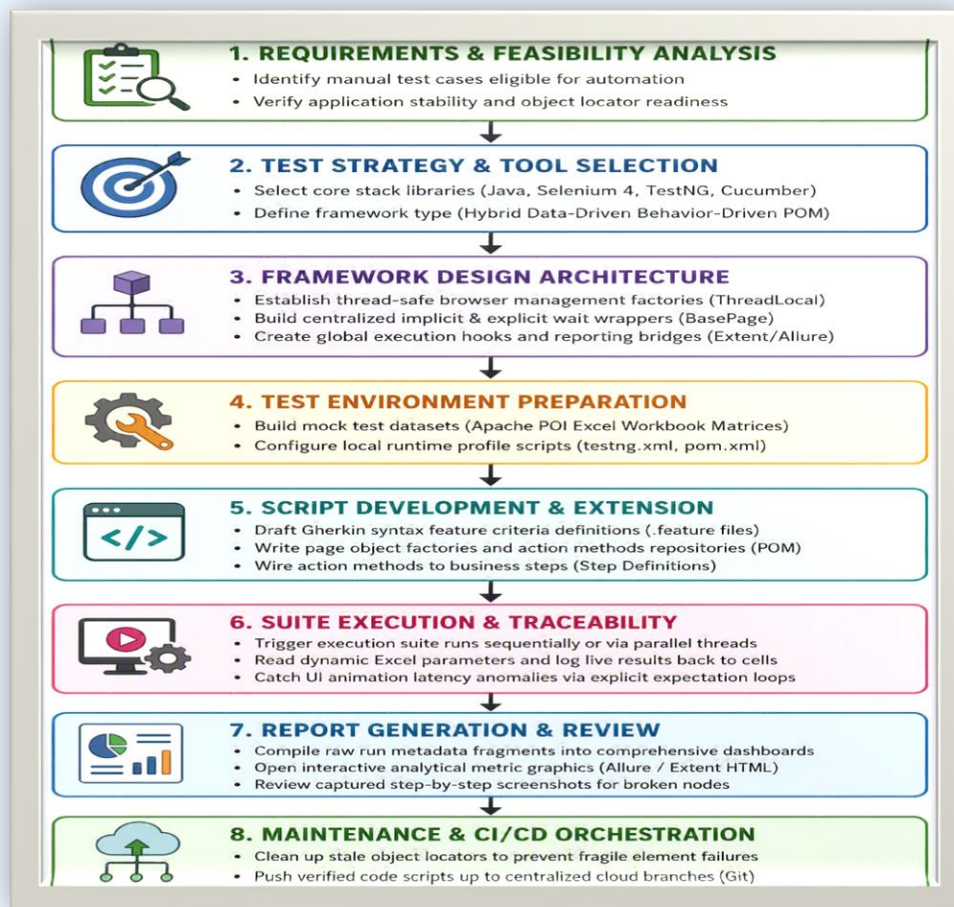


Figure 2. Automation Testing Process

1.2 PROBLEM STATEMENT

Manual testing of e-commerce applications presents several challenges:

- High execution time
- Repetitive test execution
- Increased human errors
- Difficult maintenance of test cases
- Limited test coverage

Automation testing addresses these issues through reusable and scalable test scripts.

1.3 OBJECTIVES

The primary objectives of this project are:

- Automate end-to-end e-commerce workflows
- Minimize manual testing effort
- Increase test coverage
- Improve execution efficiency
- Generate comprehensive reports
- Implement data-driven testing
- Follow industry-standard automation practices

CHAPTER 2: LITERATURE REVIEW

2.1 SELENIUM WEBDRIVER

Selenium WebDriver is an open-source automation tool used for automating web browsers. It supports multiple browsers and programming languages, making it one of the most popular choices for web application testing.

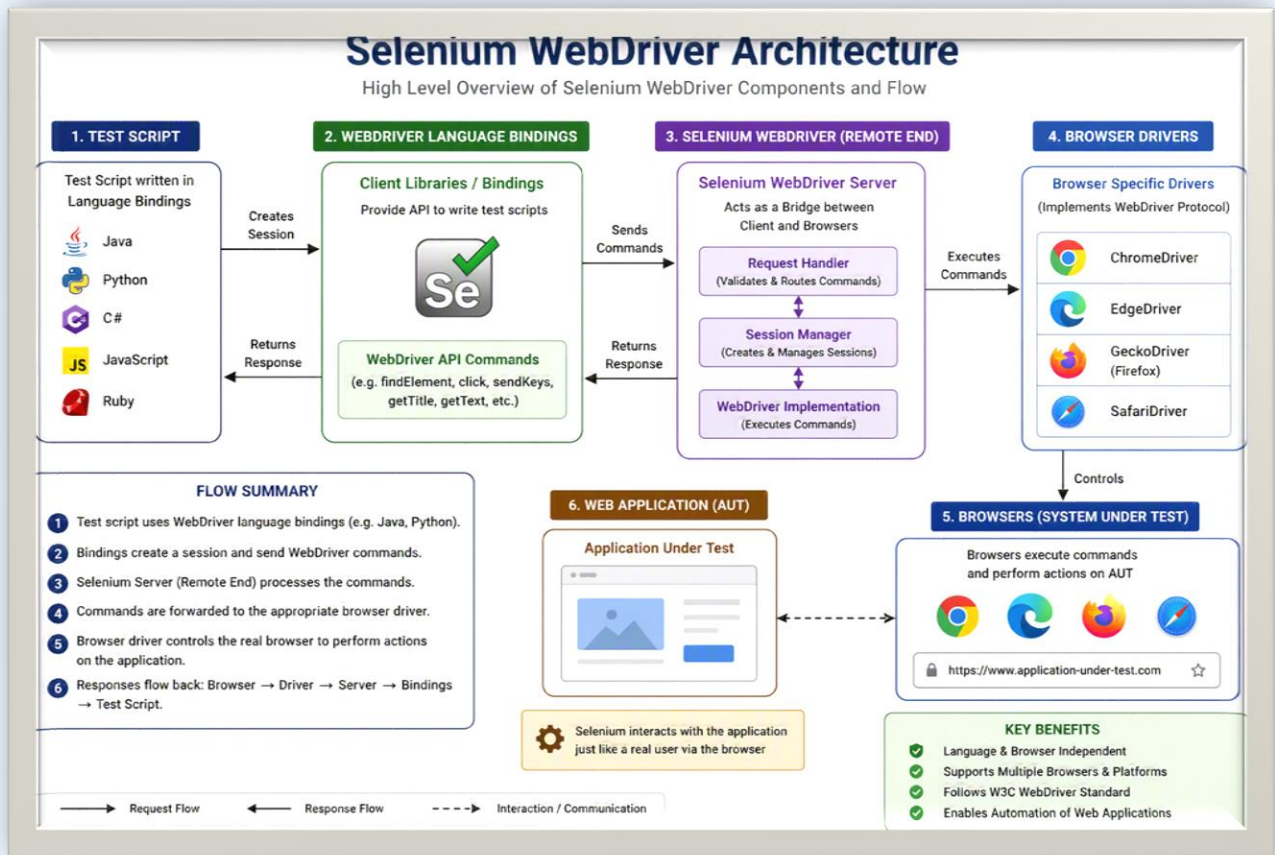


Figure 3. Selenium WebDriver Architecture

2.2 CUCUMBER BDD

Cucumber enables Behavior Driven Development by allowing stakeholders to understand test scenarios written in plain English using Gherkin syntax.

Example:

Cucumber

Scenario: User Login

Given the user is on the login page

When the user enters valid credentials

Then the user should be logged in successfully

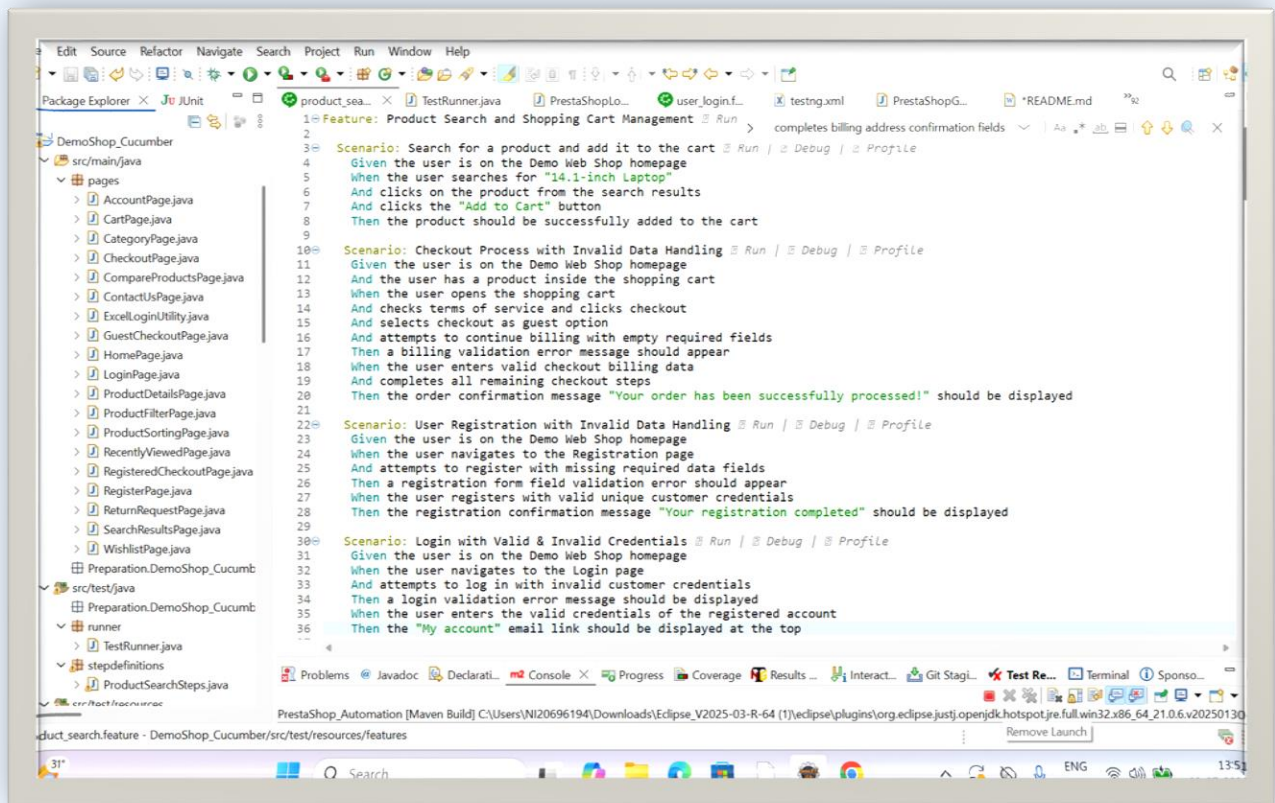


Figure 4. Sample Cucumber Feature File

2.3 TestNG

TestNG is a testing framework used for organizing and executing test cases. It provides annotations, data providers, assertions, and report generation capabilities.

CHAPTER 3: SYSTEM DESIGN

3.1 FRAMEWORK ARCHITECTURE

The project follows the Page Object Model (POM).

The architecture separates test scripts from page elements and business logic, improving maintainability.

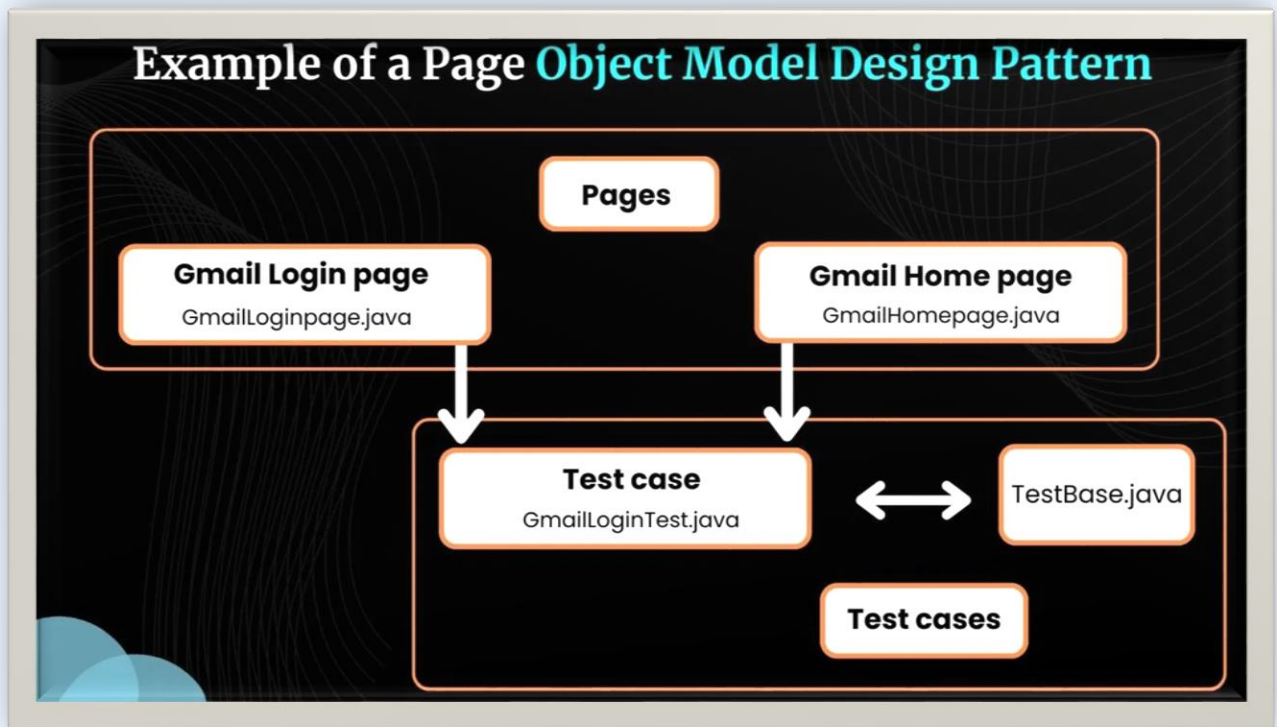


Figure 5. Framework Architecture

3.2 PROJECT STRUCTURE

The framework structure is organized as follows:

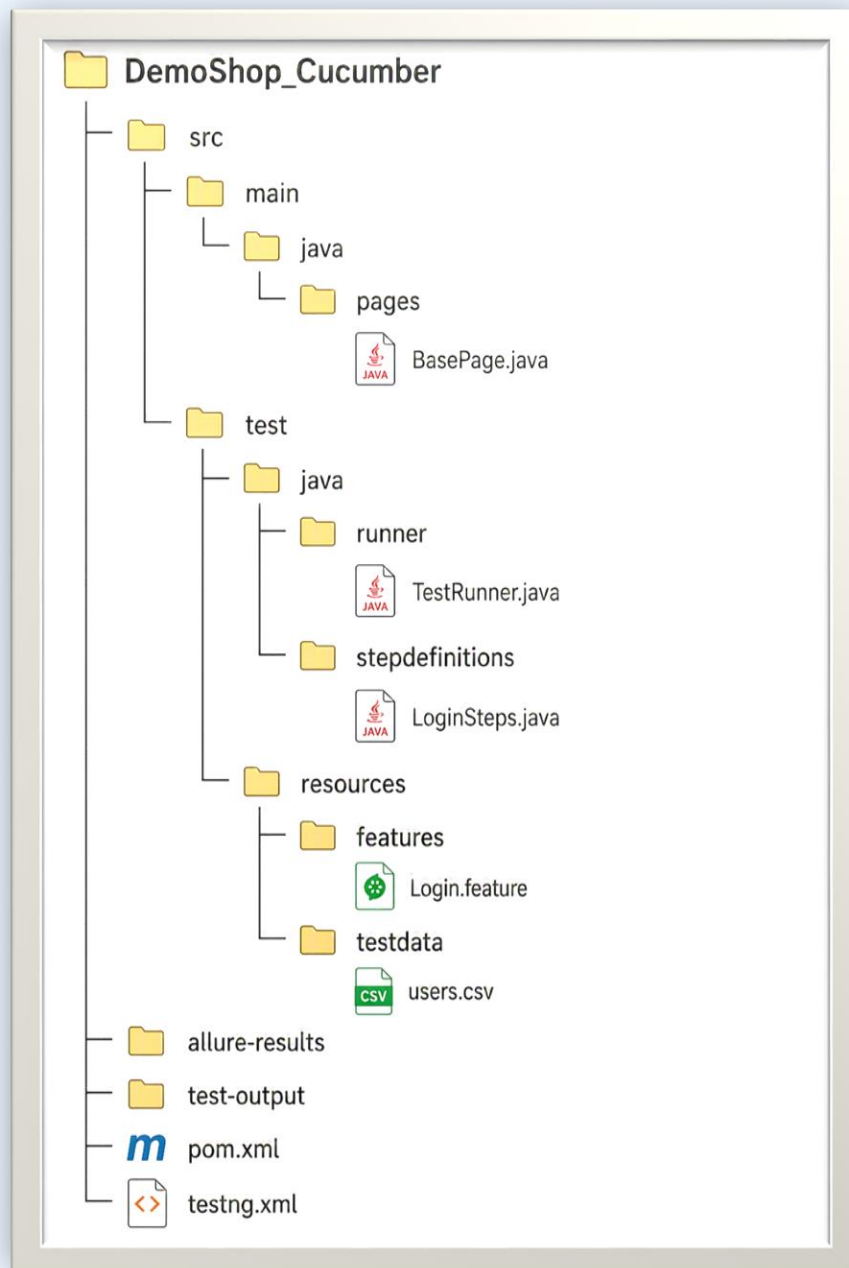


Figure 6. Project Structure in Eclipse

CHAPTER 4: TECHNOLOGIES USED

Technology	Purpose
Java	Programming Language
Selenium WebDriver	Browser Automation

Cucumber	BDD Testing
TestNG	Test Execution
Maven	Build Management
Apache POI	Excel Handling
Allure Report	Reporting
Extent Report	Reporting
GitHub	Version Control

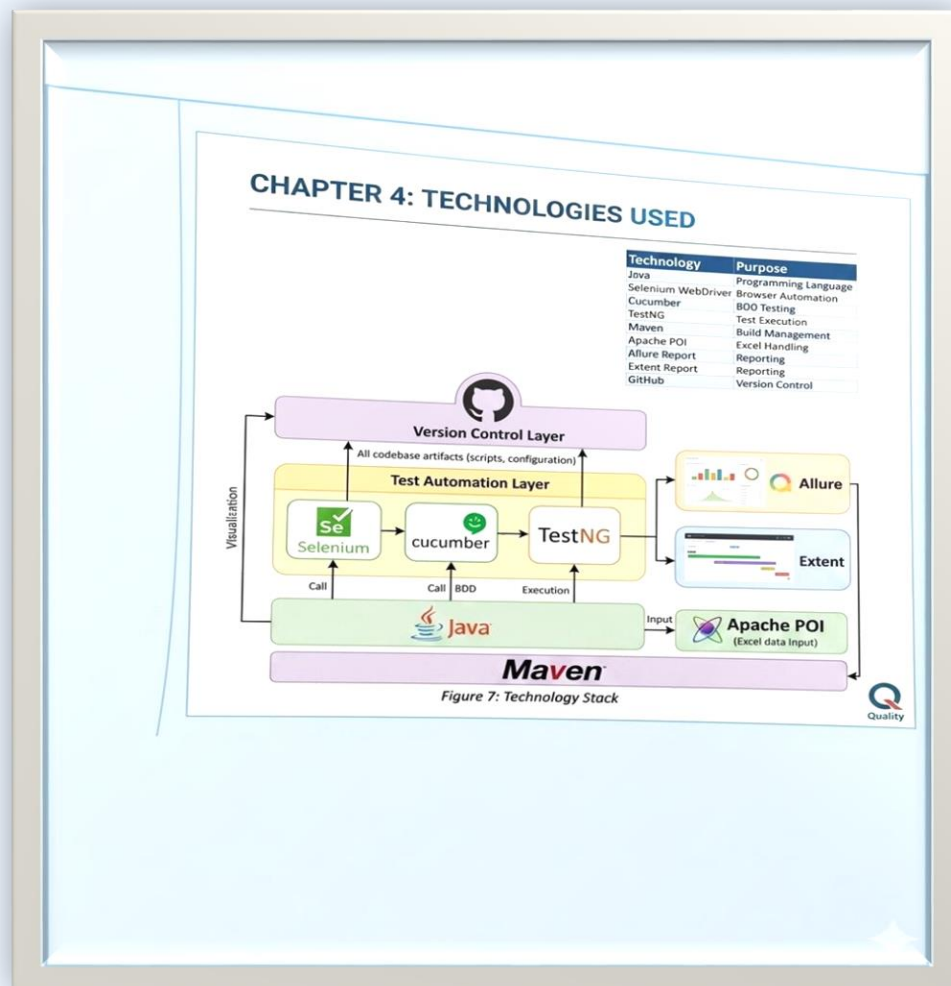


Figure 7. Technology Stack

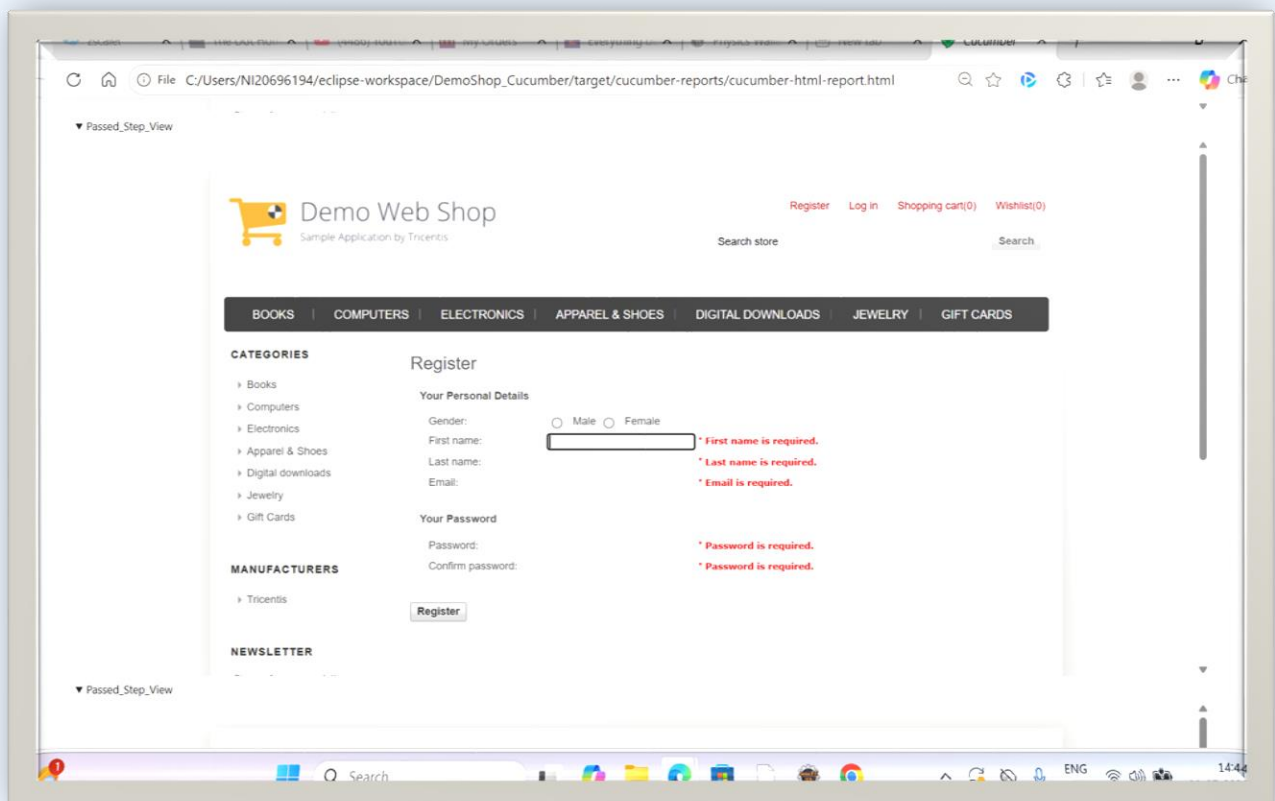
CHAPTER 5: IMPLEMENTATION

5.1 USER REGISTRATION MODULE

The registration module validates the creation of a new user account by entering personal and login information.

Steps

1. Navigate to Registration Page
2. Enter User Details
3. Submit Registration Form
4. Verify Registration Success



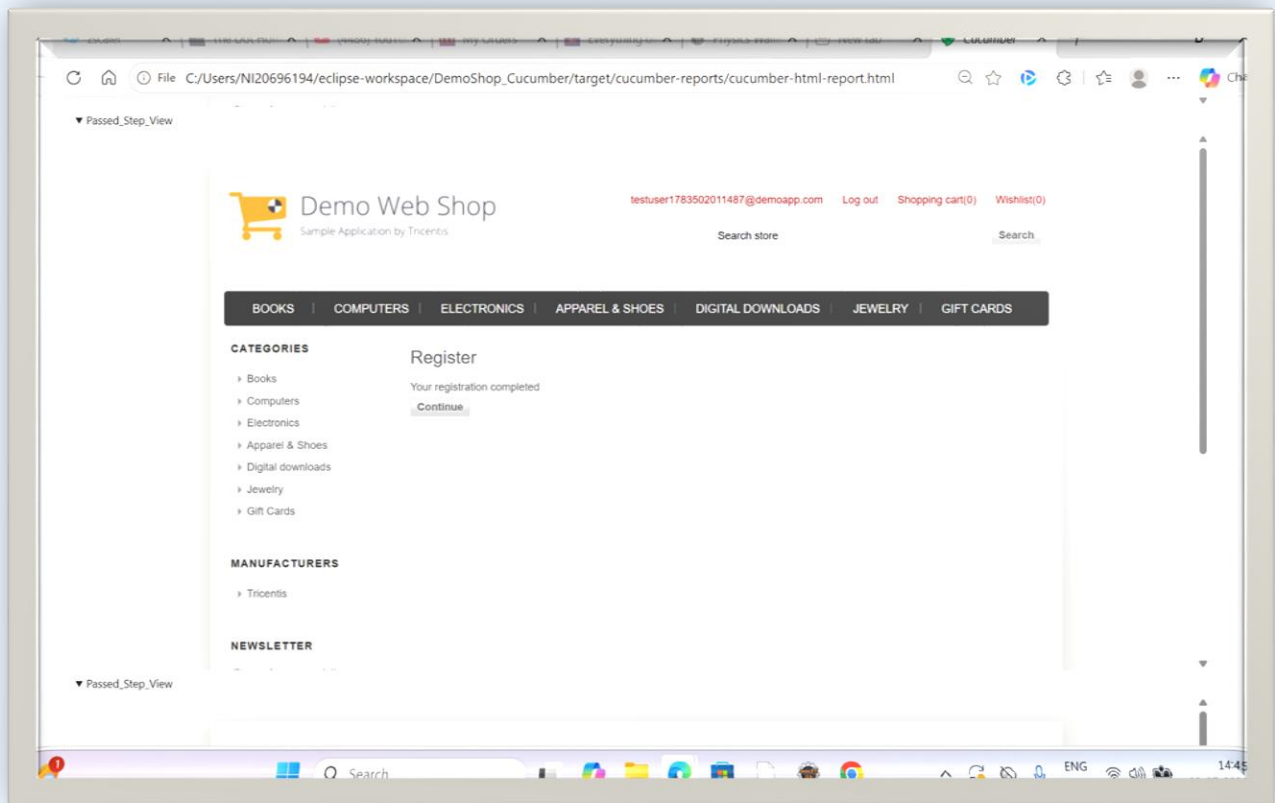


Figure 8. User Registration Page

5.2 LOGIN MODULE

The login module verifies authentication functionality with valid credentials.

Steps

1. Open Login Page
2. Enter Email
3. Enter Password
4. Click Login
5. Verify Successful Login

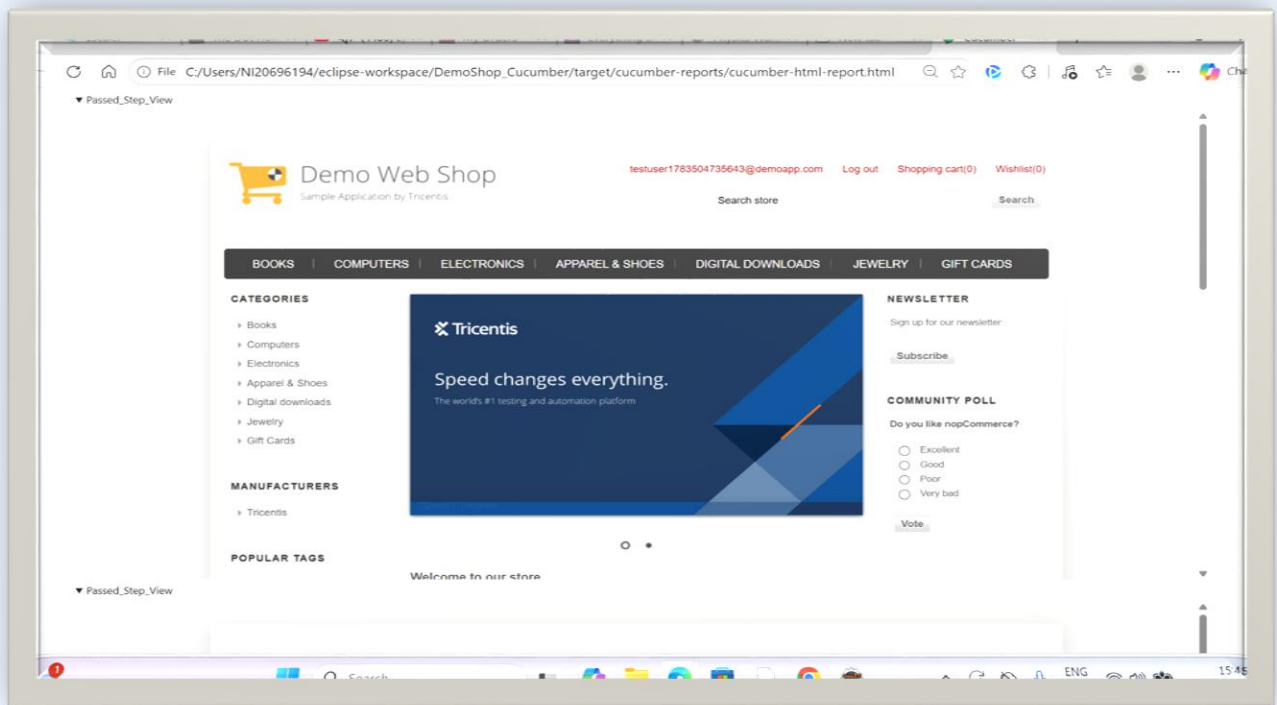
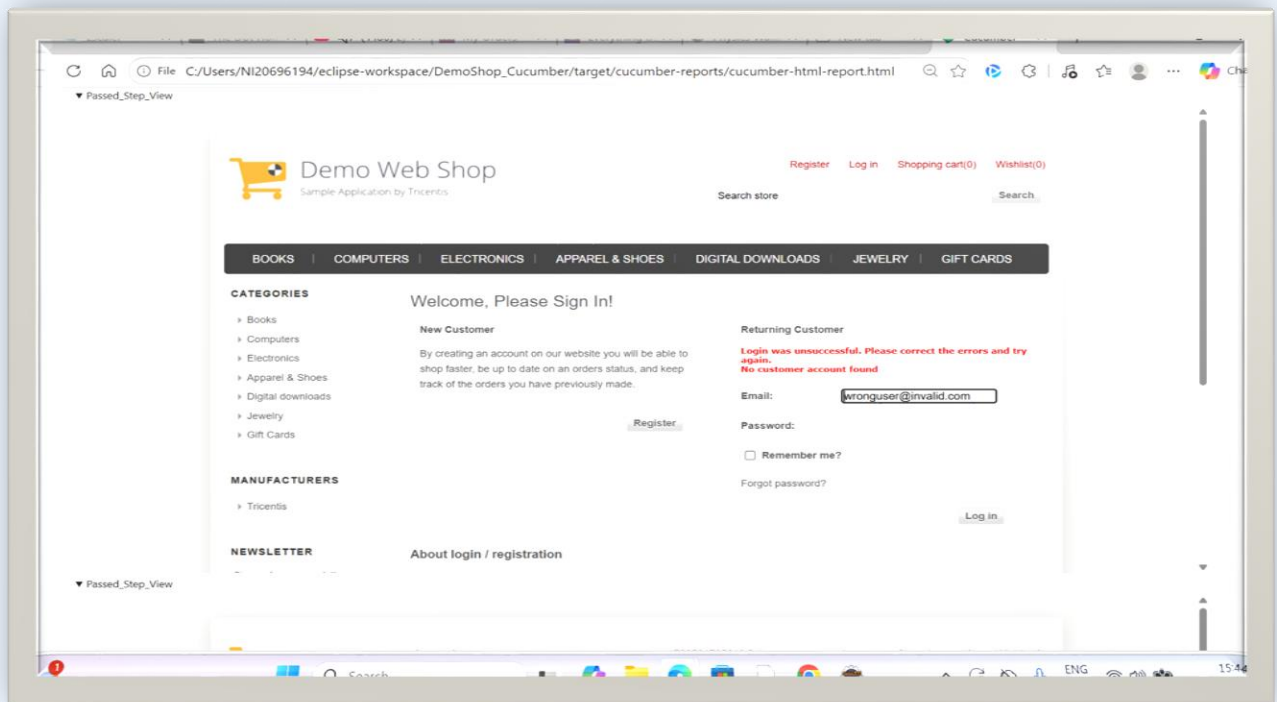


Figure 9. Login Functionality

5.3 PRODUCT SEARCH MODULE

Users can search for products using keywords.

Validation

- Search execution
- Product visibility
- Search accuracy

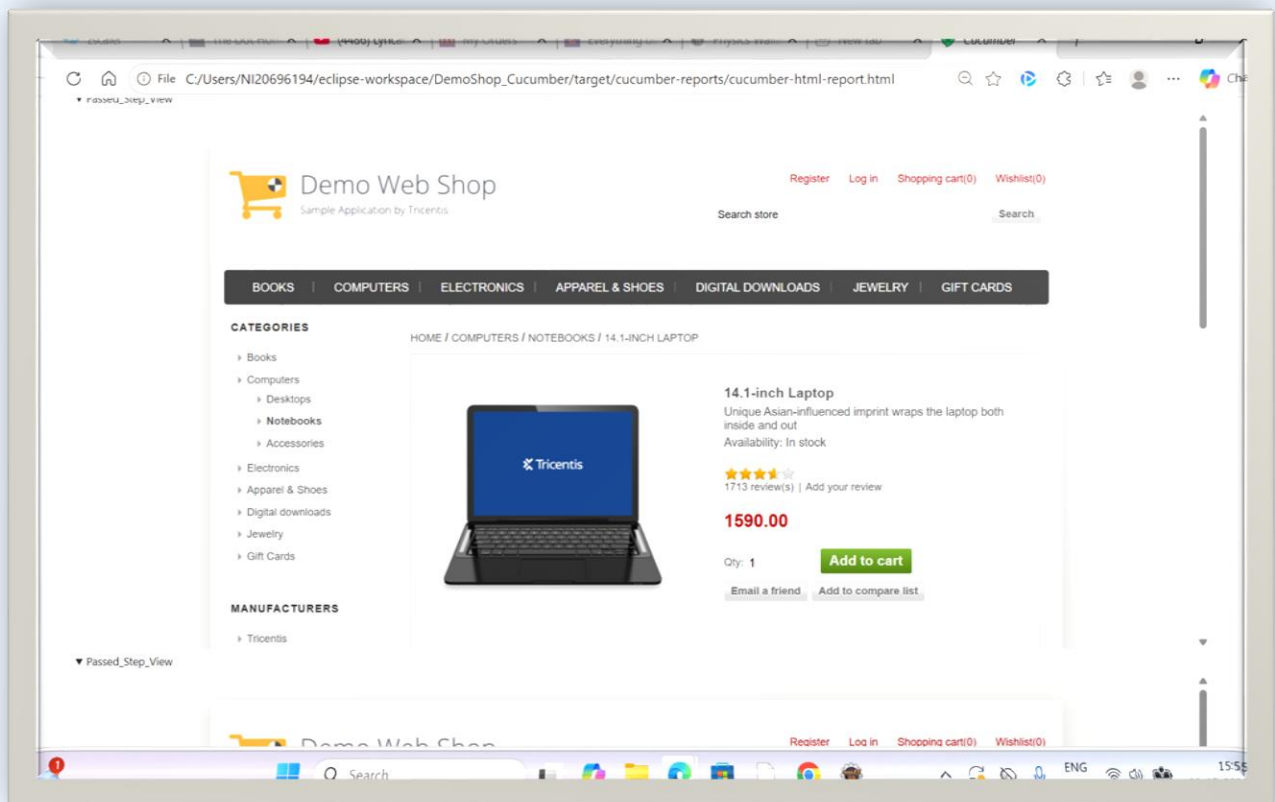


Figure 10. Product Search Results

5.4 PRODUCT FILTER MODULE

The filter module validates category-based filtering.

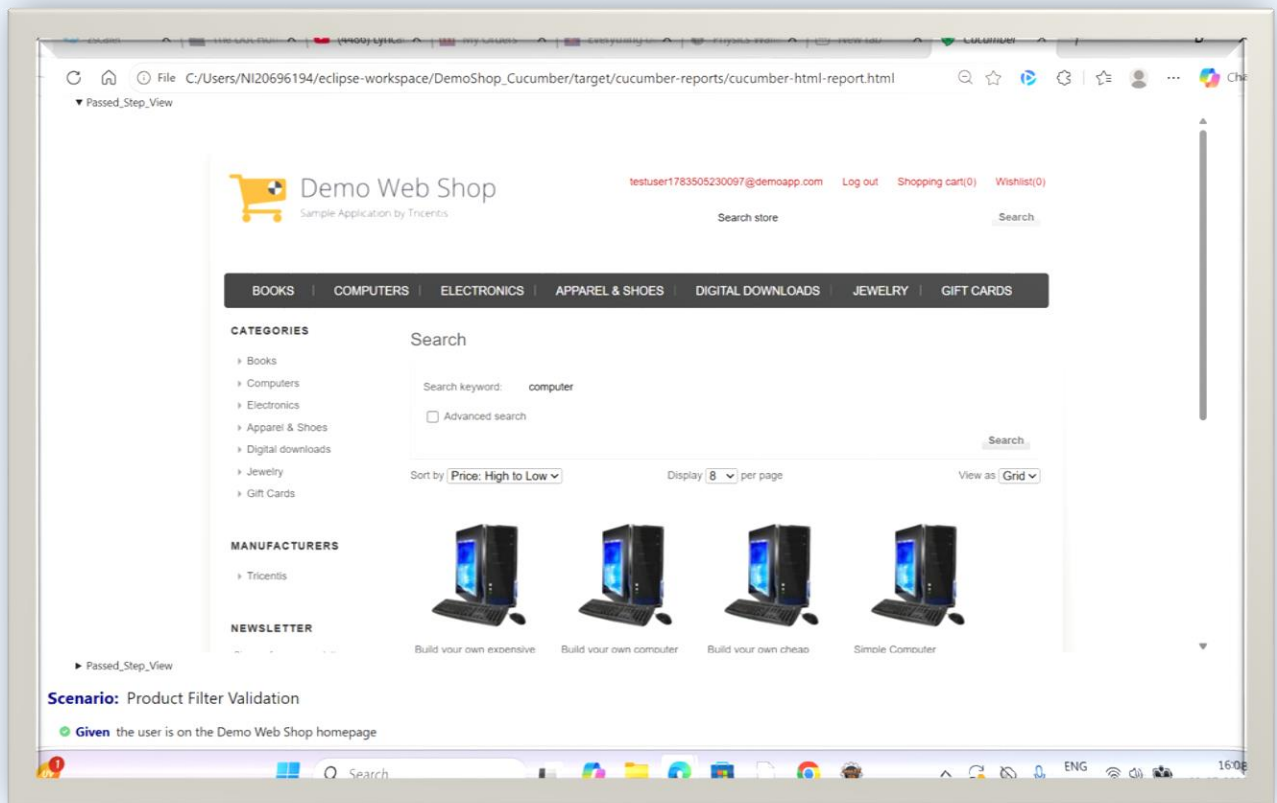


Figure 11. Product Filtering Feature

5.5 COMPARE PRODUCTS MODULE

The compare functionality allows users to compare multiple products before purchasing.

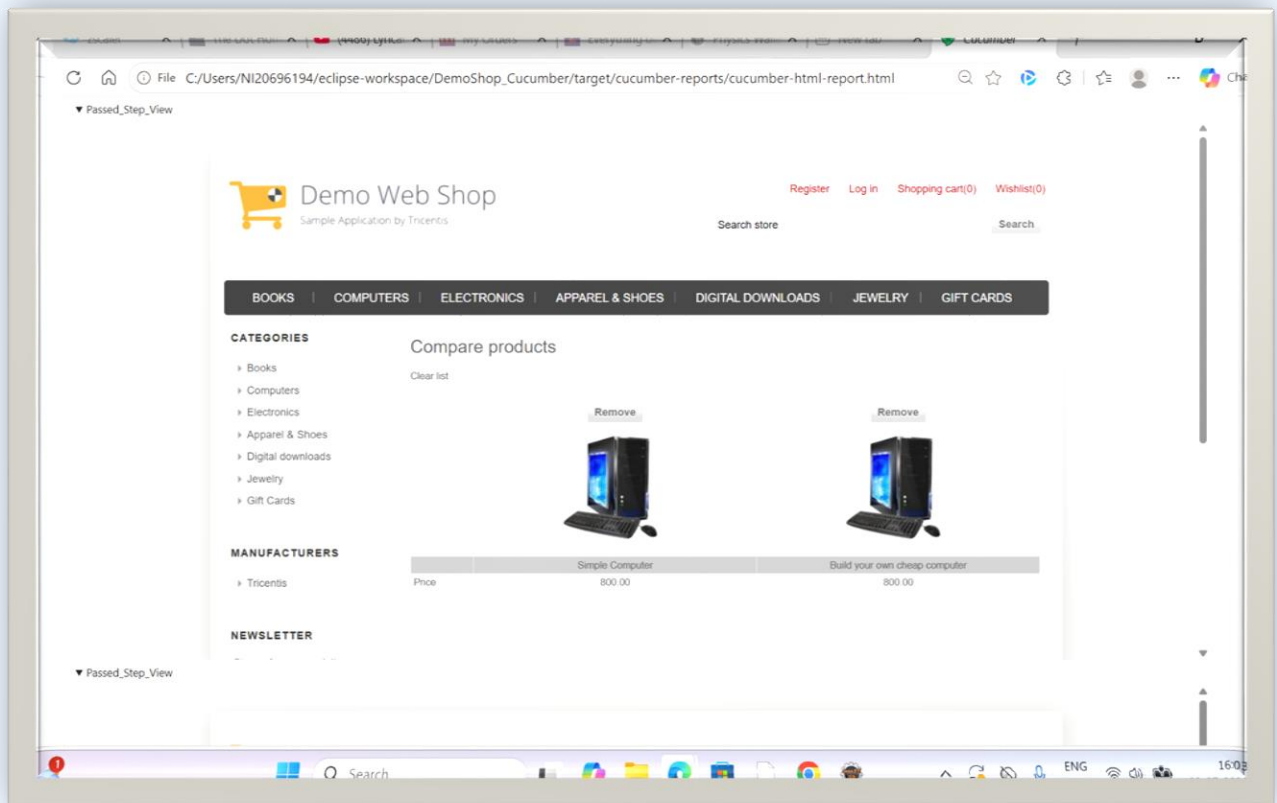


Figure 12. Product Comparison Screen

5.6 WISHLIST MODULE

The wishlist module stores favorite items for future purchases.

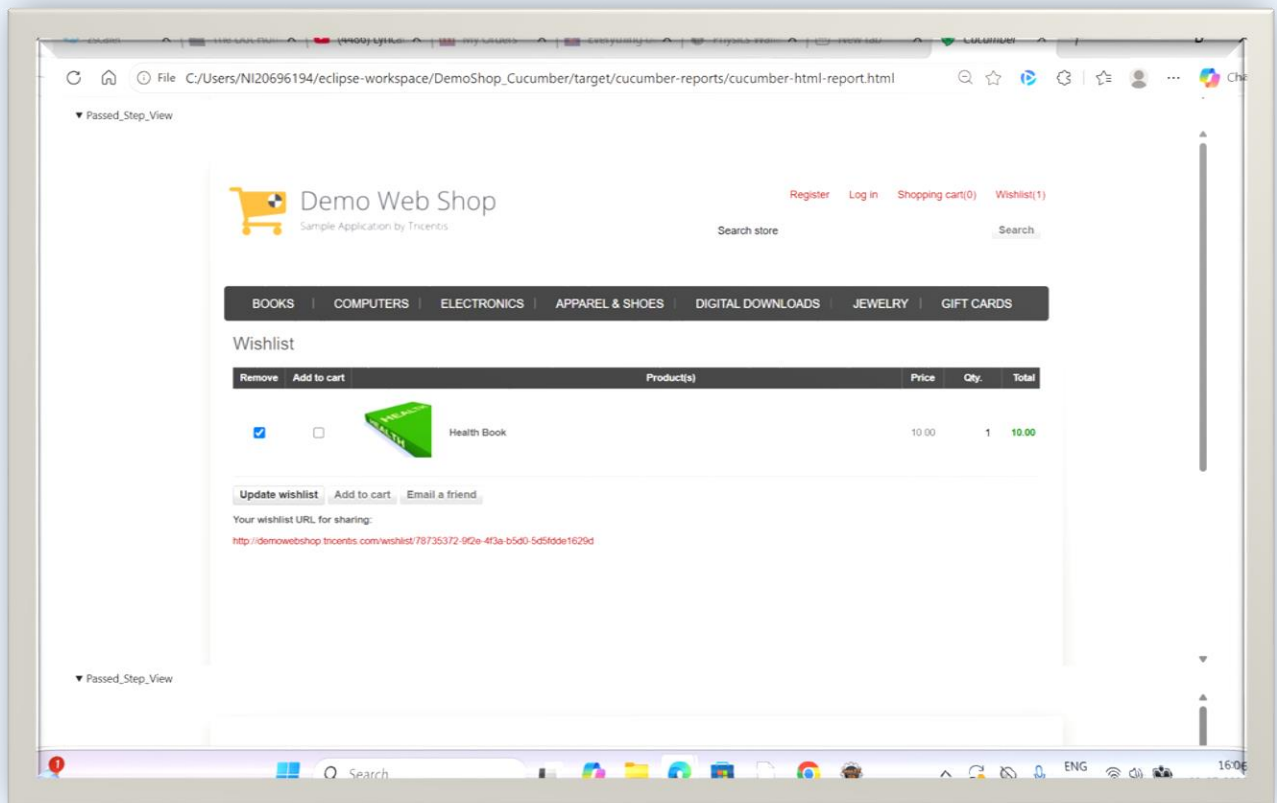


Figure 13. Wishlist Management

5.7 SHOPPING CART MODULE

Cart functionality includes:

- Add Product
- Remove Product

- Update Quantity
- Verify Total

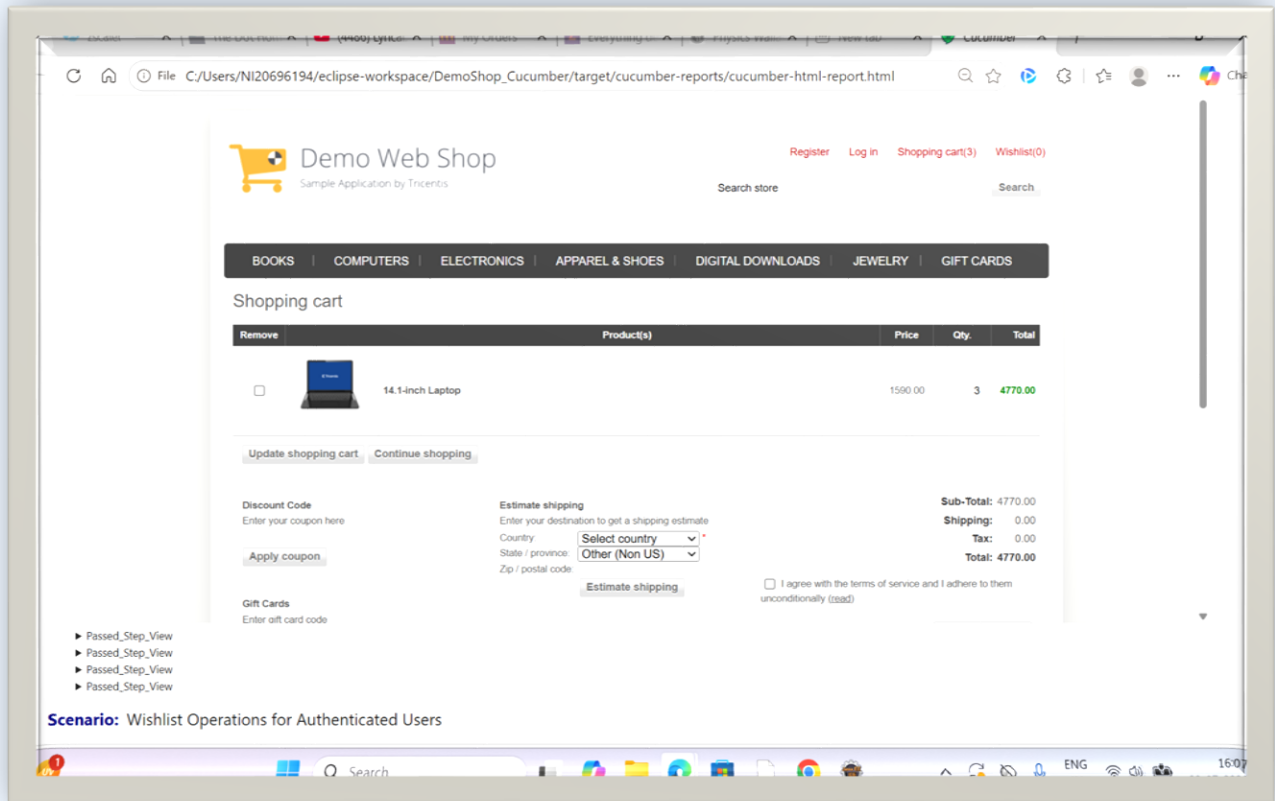


Figure 14. Shopping Cart Operations

CHAPTER 6: CHECKOUT PROCESS

6.1 GUEST CHECKOUT

Guest users can place orders without registration.

Workflow

- Product Selection
- Cart Review
- Billing Information

- Shipping Method
- Payment Method
- Order Confirmation

The screenshot displays a web browser window with the URL `C:/Users/NI20696194/eclipse-workspace/DemoShop_Cucumber/target/cucumber-reports/cucumber-html-report.html`. The page is titled "Demo Web Shop" and is a "Sample Application by Tricentis". It features a navigation bar with categories: BOOKS, COMPUTERS, ELECTRONICS, APPAREL & SHOES, DIGITAL DOWNLOADS, JEWELRY, and GIFT CARDS. The main content area is titled "Checkout" and contains a "Billing Address" form. The form fields include: First name, Last name, Email, Company, Country (a dropdown menu currently showing "Select country"), State / province (a dropdown menu currently showing "Other (Non US)"), City, Address 1, Address 2, Zip / postal code, Phone number, and Fax number. Each field has a red asterisk indicating it is required. Below the form is a "Continue" button. The browser's address bar and the Windows taskbar are visible at the bottom.

Figure 15. Guest Checkout Details

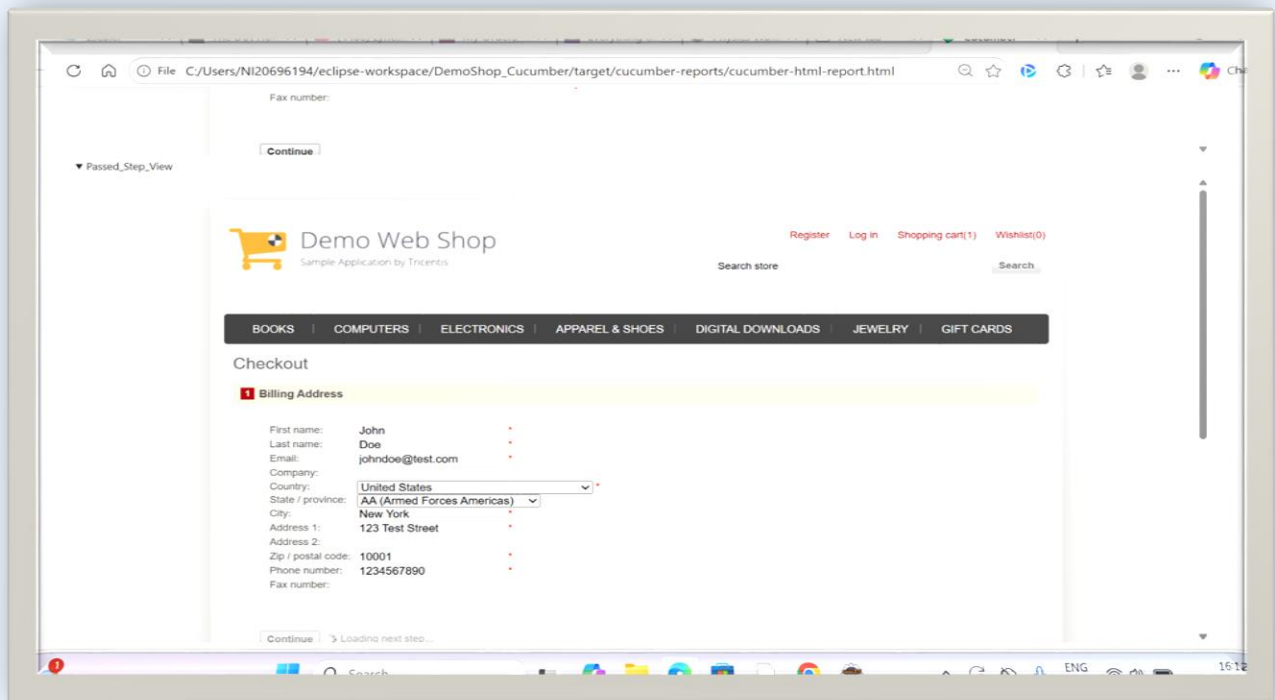


Figure 16. Billing Address Entry

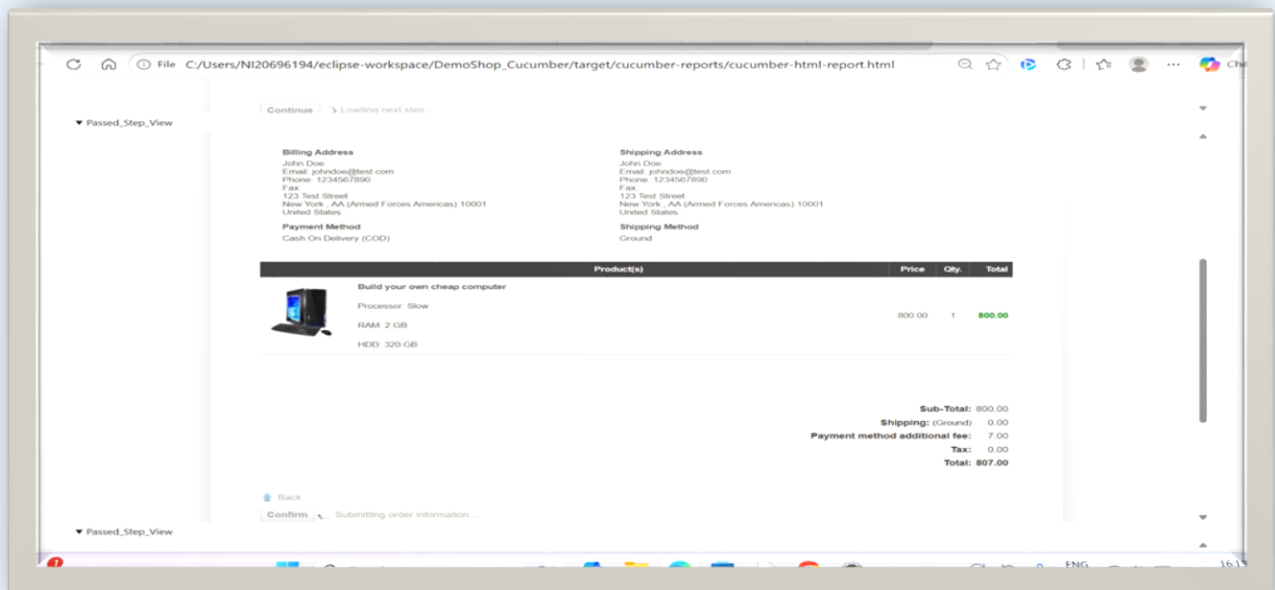


Figure 17. Shipping Method Selection and Payment Method Selection

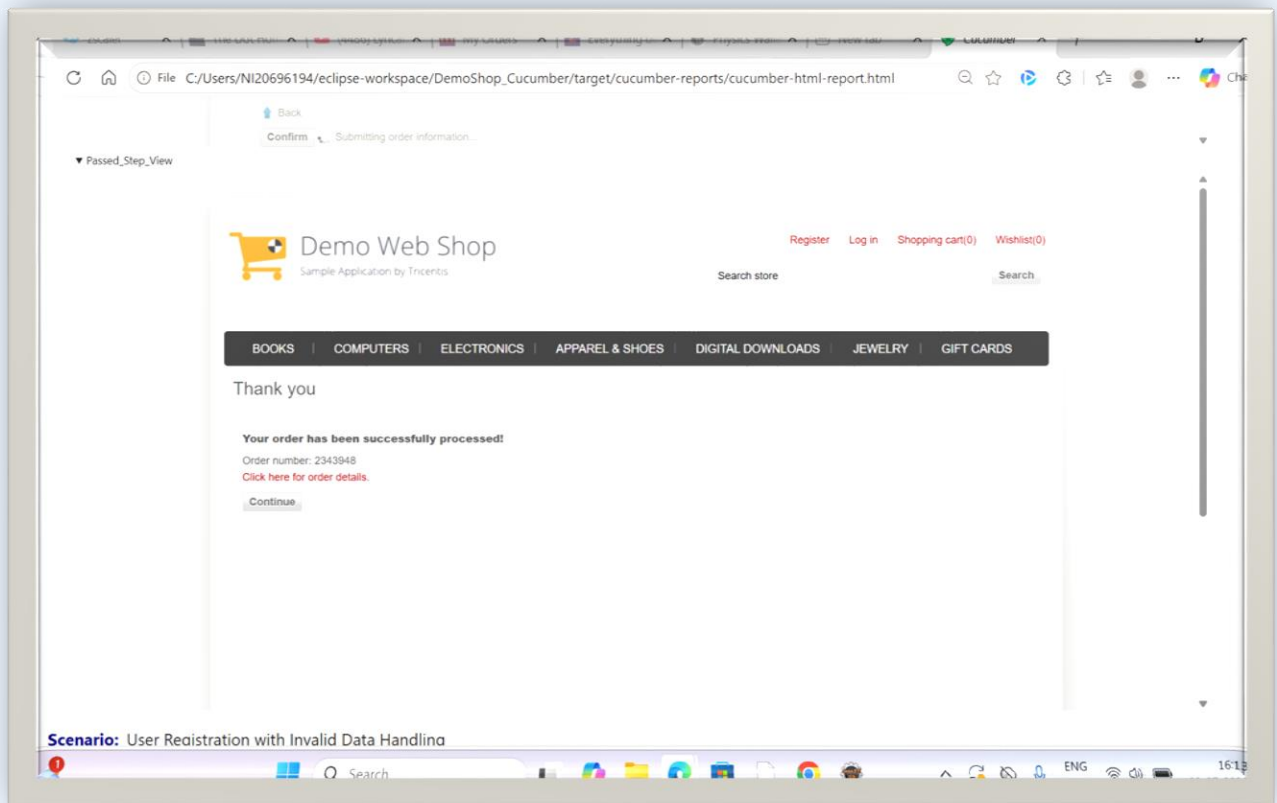


Figure 18. Order Confirmation

6.2 REGISTERED USER CHECKOUT

The registered checkout process validates complete order placement for authenticated users.

The screenshot displays a web browser window with the following elements:

- Address Bar:** File C:/Users/NI20696194/eclipse-workspace/DemoShop_Cucumber/target/cucumber-reports/cucumber-html-report.html
- Left Sidebar:**
 - CATEGORIES:** Books, Computers, Electronics, Apparel & Shoes, Digital downloads, Jewelry, Gift Cards.
 - MANUFACTURERS:** Tricents.
 - NEWSLETTER:** About login / registration.
- Main Content Area:**
 - Welcome, Please Sign In!**
 - Checkout as a guest or register**
 - Register and save time!**
 - Register with us for future convenience:
 - Fast and easy check out
 - Easy access to your order history and status
 - Returning Customer:** Email: [input field], Password: [input field], ☐ Remember me?, [Forgot password?](#)
 - [Checkout as Guest](#), [Register](#), [Log in](#)
 - Billing Address:** Ankit Surthar, Email: checkoutuser1783505069967@demoapp.com, Phone: 9876543210, Fax: 456 Automation Way, Chicago, AA (Armed Forces Americas) 60601, United States.
 - Shipping Address:** Ankit Surthar, Email: checkoutuser1783505069967@demoapp.com, Phone: 9876543210, Fax: 456 Automation Way, Chicago, AA (Armed Forces Americas) 60601, United States.
 - Payment Method:** Cash On Delivery (COD)
 - Shipping Method:** Ground
- Product Table:**

Product(s)	Price	Qty.	Total
Simple Computer Processor: Slow RAM: 2 GB HDD: 320 GB	800.00	1	800.00

Figure 19. Registered User Checkout

CHAPTER 7: CUSTOMER SERVICES MODULES

7.1 RETURN REQUEST MODULE

The return request module allows customers to request product returns.

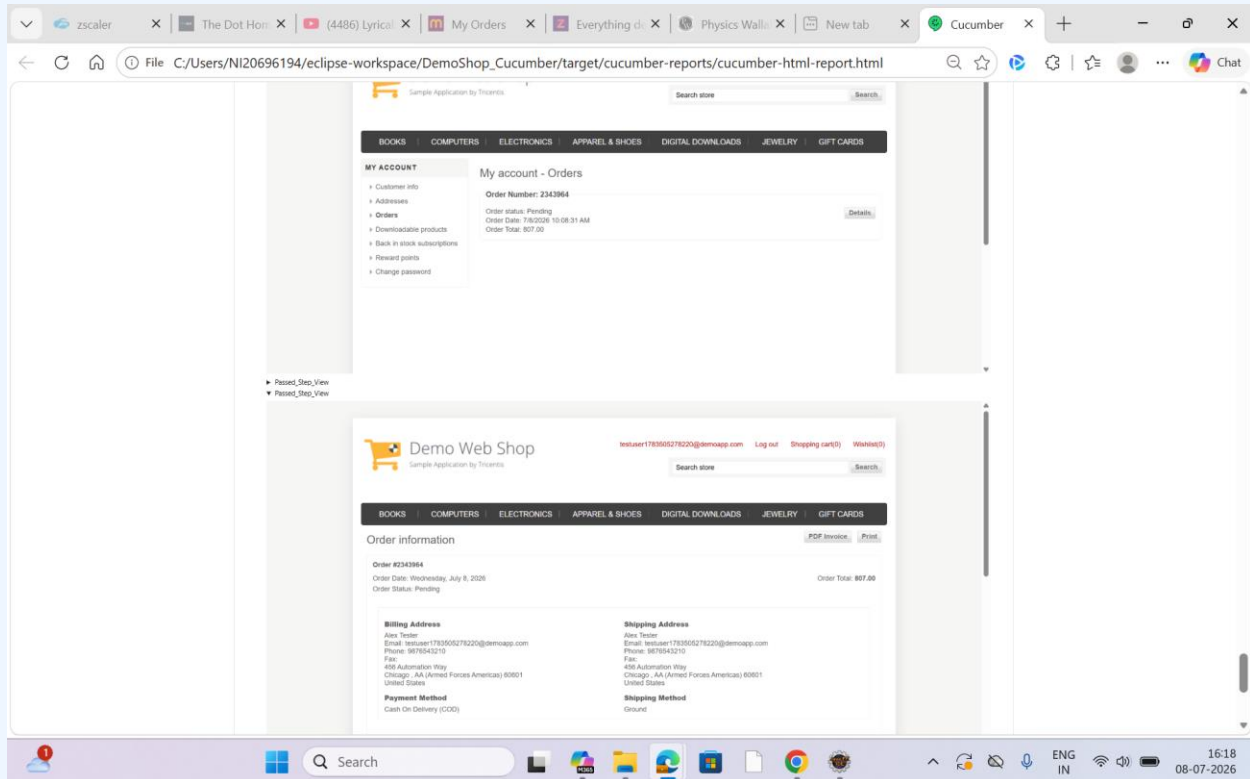


Figure 20. Return Request

7.2 RECENTLY VIEWED PRODUCTS

This feature allows customers to view previously visited products.

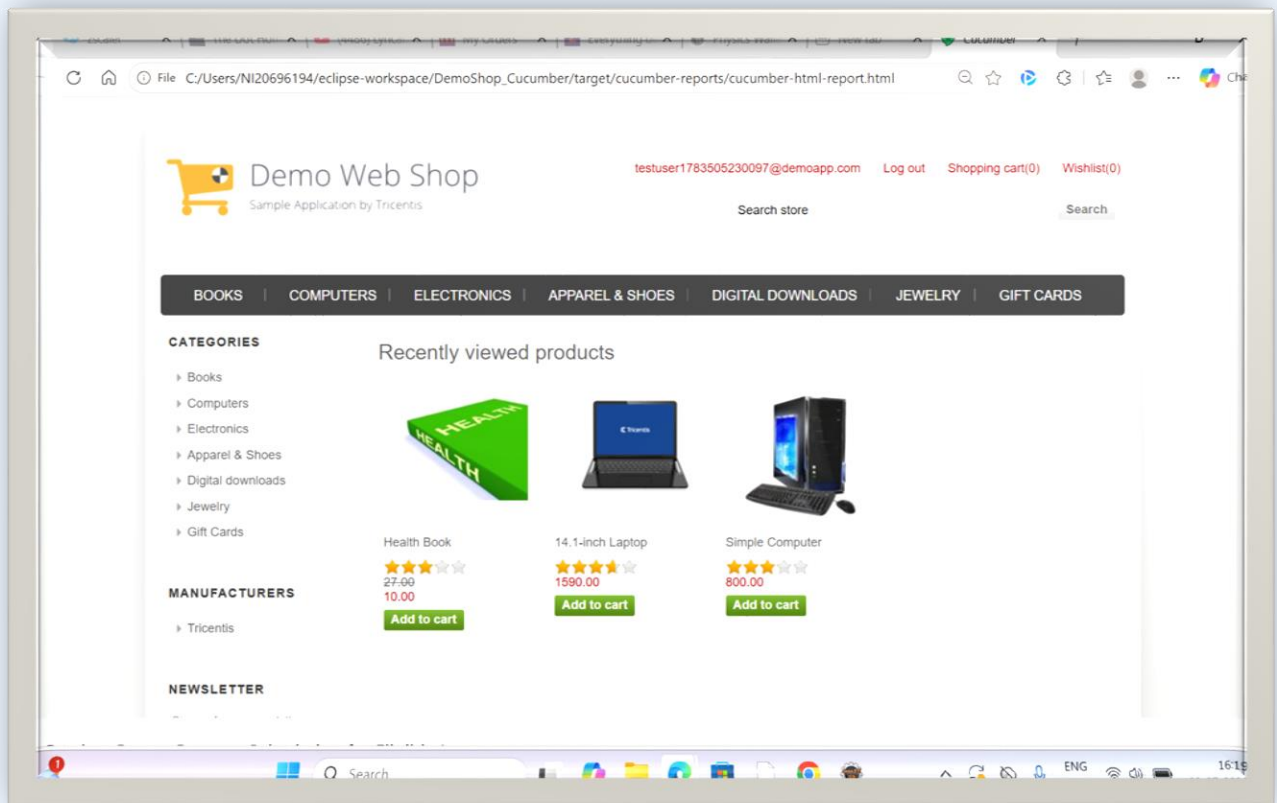


Figure 21. Recently Viewed Products

CHAPTER 8: DATA DRIVEN TESTING

The framework uses Excel files for test data management.

Benefits

- Externalized Data
- Reusable Data Sets

- Improved Maintainability

Files Used

- UserCredentials.xlsx
- ExcelLoginUtility.java

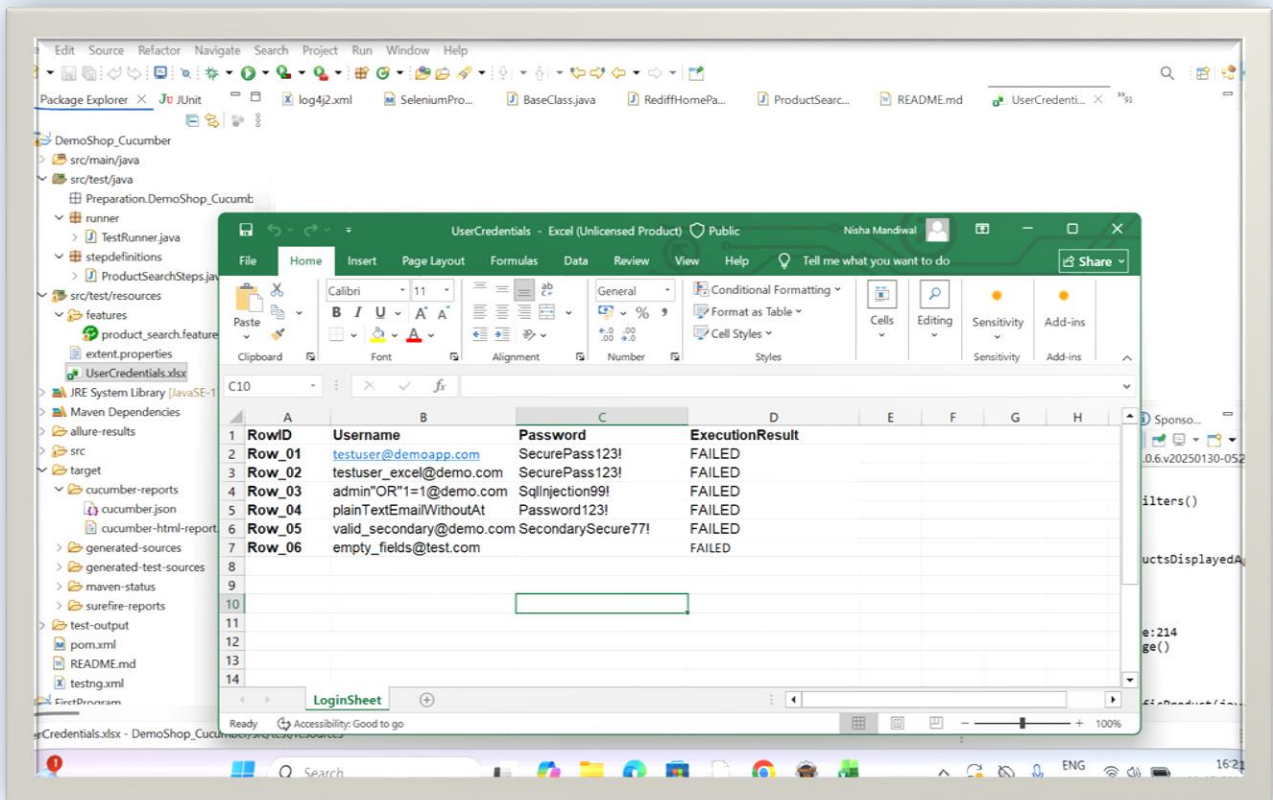


Figure 22. Excel-Based Test Data

CHAPTER 9: TEST EXECUTION

The framework executes test scenarios through the Cucumber Test Runner.

Test Runner Responsibilities

- Execute Feature Files
- Configure Reports
- Manage TestNG Integration

- Control Scenario Execution

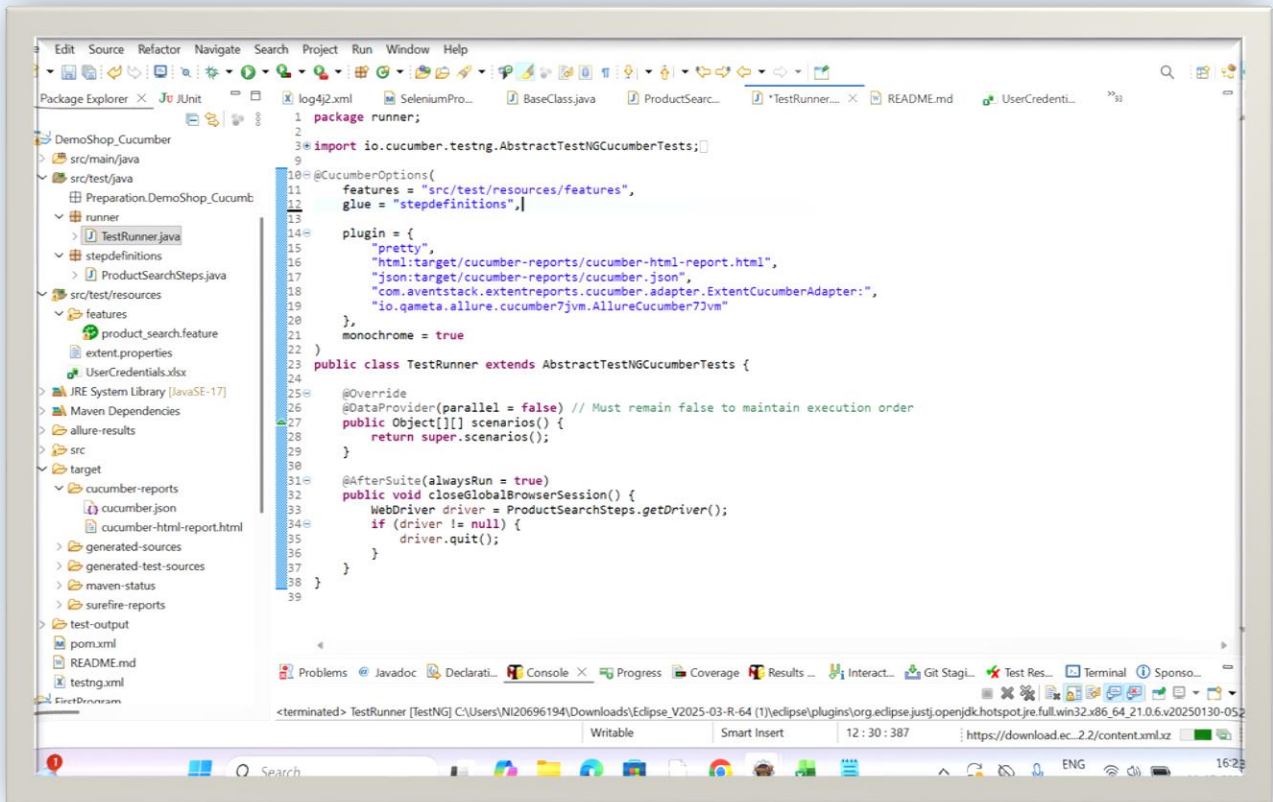


Figure 23. Cucumber Test Runner Configuration

CHAPTER 10: REPORTING

10.1 ALLURE REPORTS

Allure reports provide detailed execution of metrics.

Features include:

- Execution Summary
- Pass/Fail Statistics
- Graphical Dashboard
- Test Logs

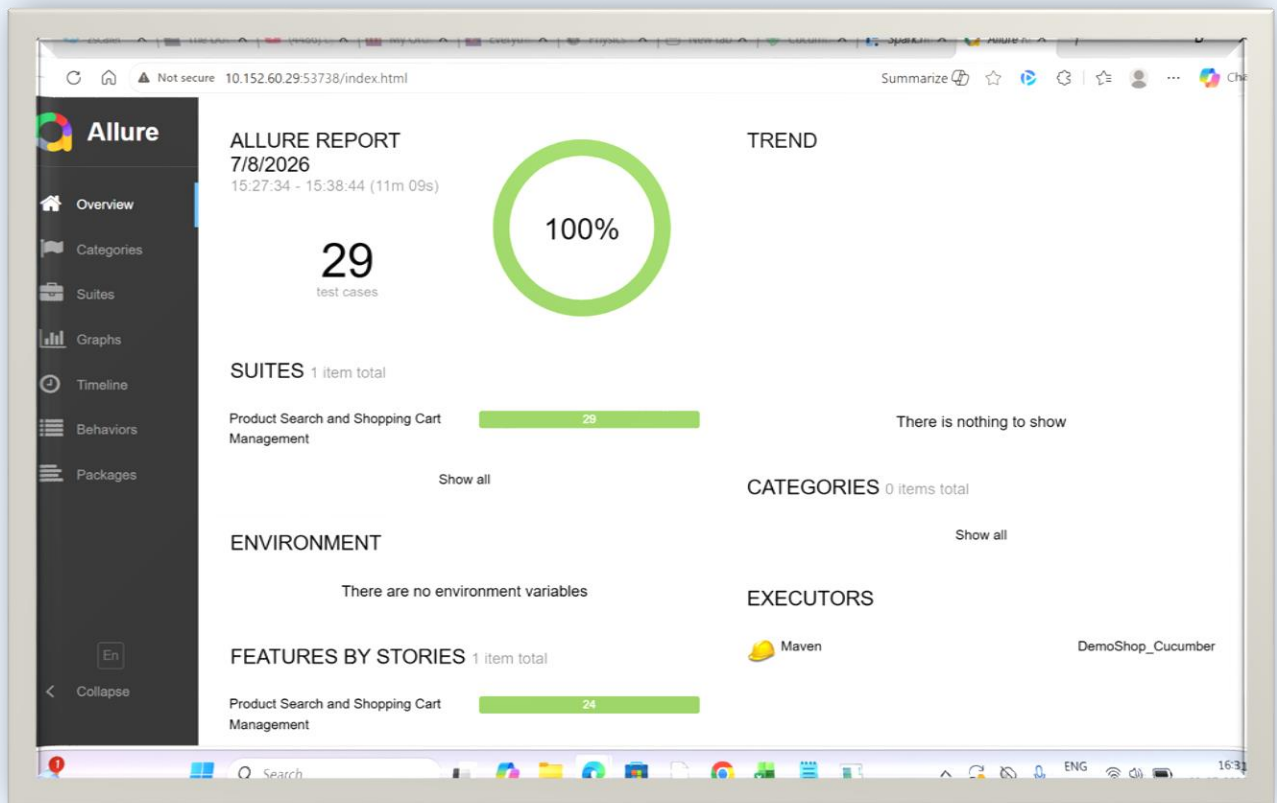


Figure 24. Allure Report Dashboard

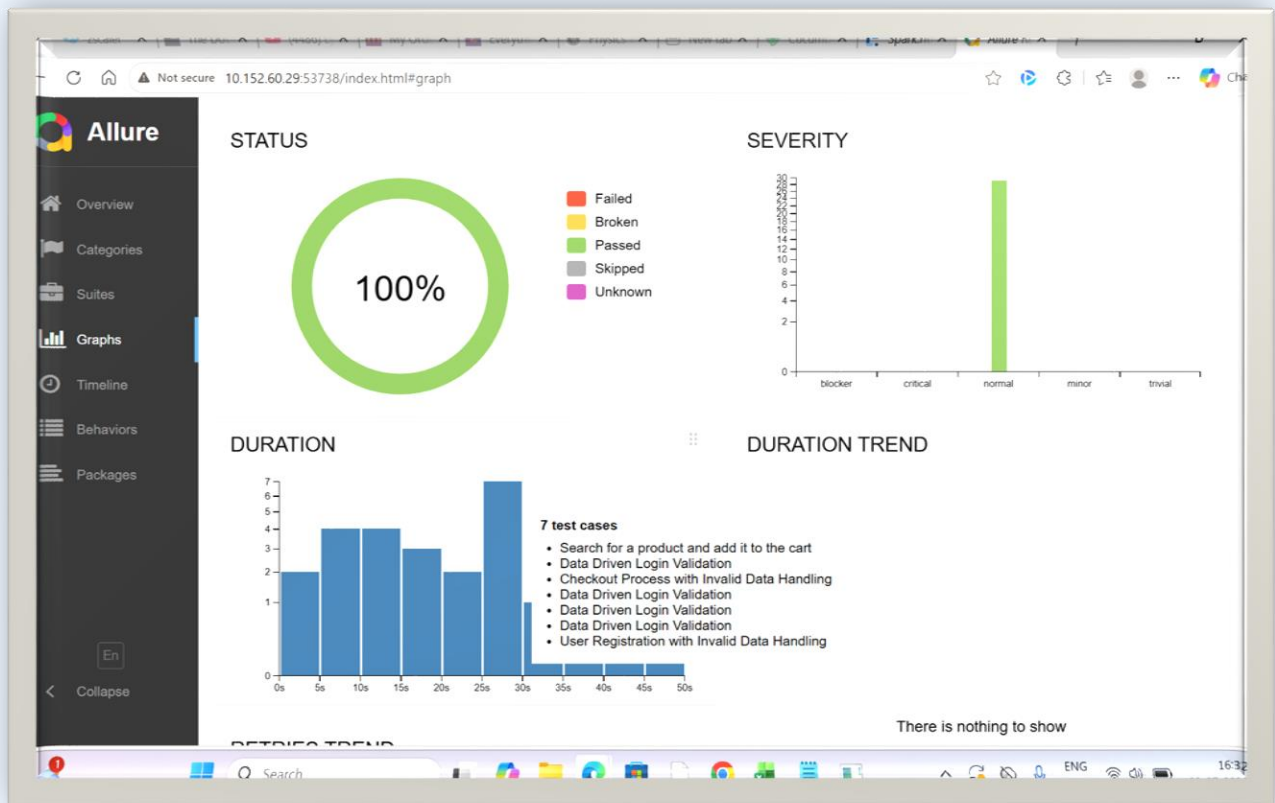


Figure 25. Allure Execution Statistics

10.2 EXTENT REPORTS

Extent Reports generate detailed HTML reports after execution.

Features include:

- Interactive Interface
- Test Logs
- Pass/Fail Status
- Screenshots

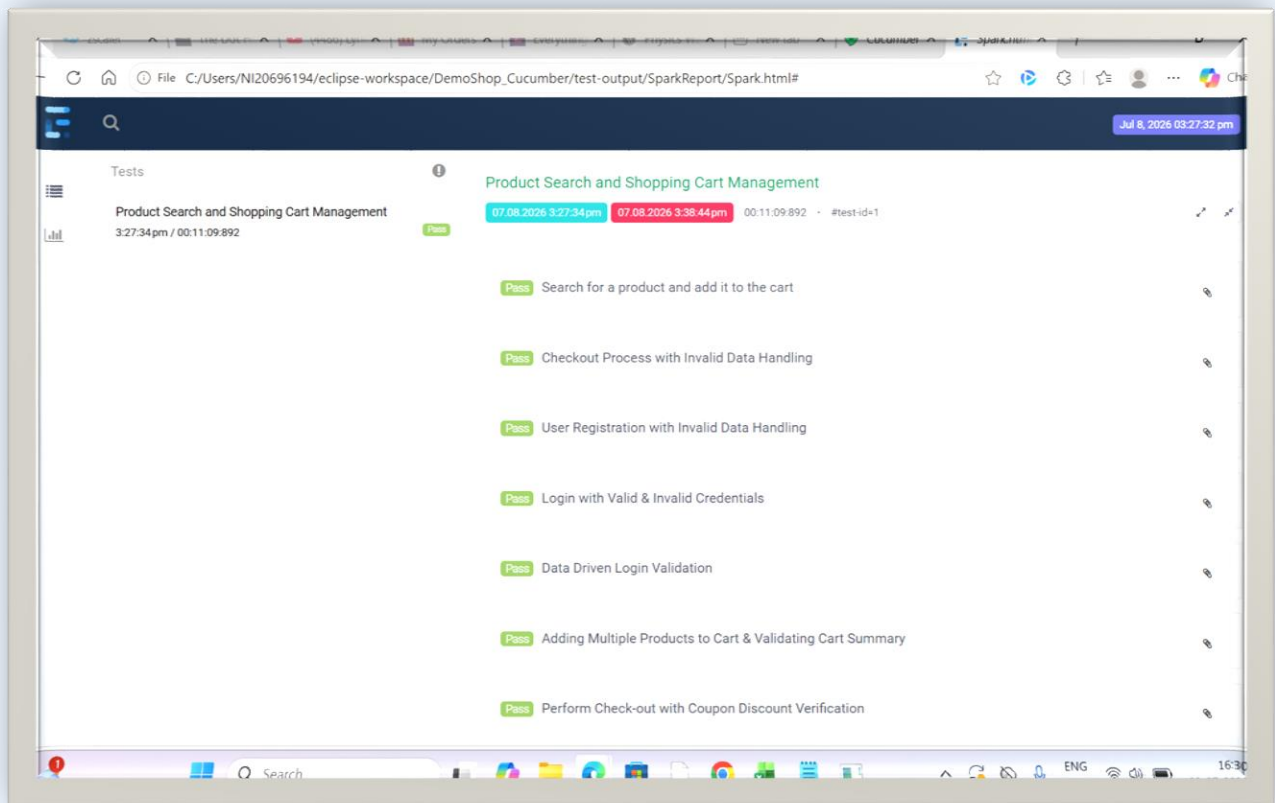


Figure 26. Extent HTML Report

10.3 BDD CUCUMBER REPORTS

BDD Cucumber Reports generate detailed execution reports based on feature files and scenarios.

Features include:

- Feature Execution Summary
- Scenario and Step Details
- Pass/Fail Status
- Error Logs
- Tag-wise Filtering
- Execution Statistics
- Screenshots
- Interactive Dashboard

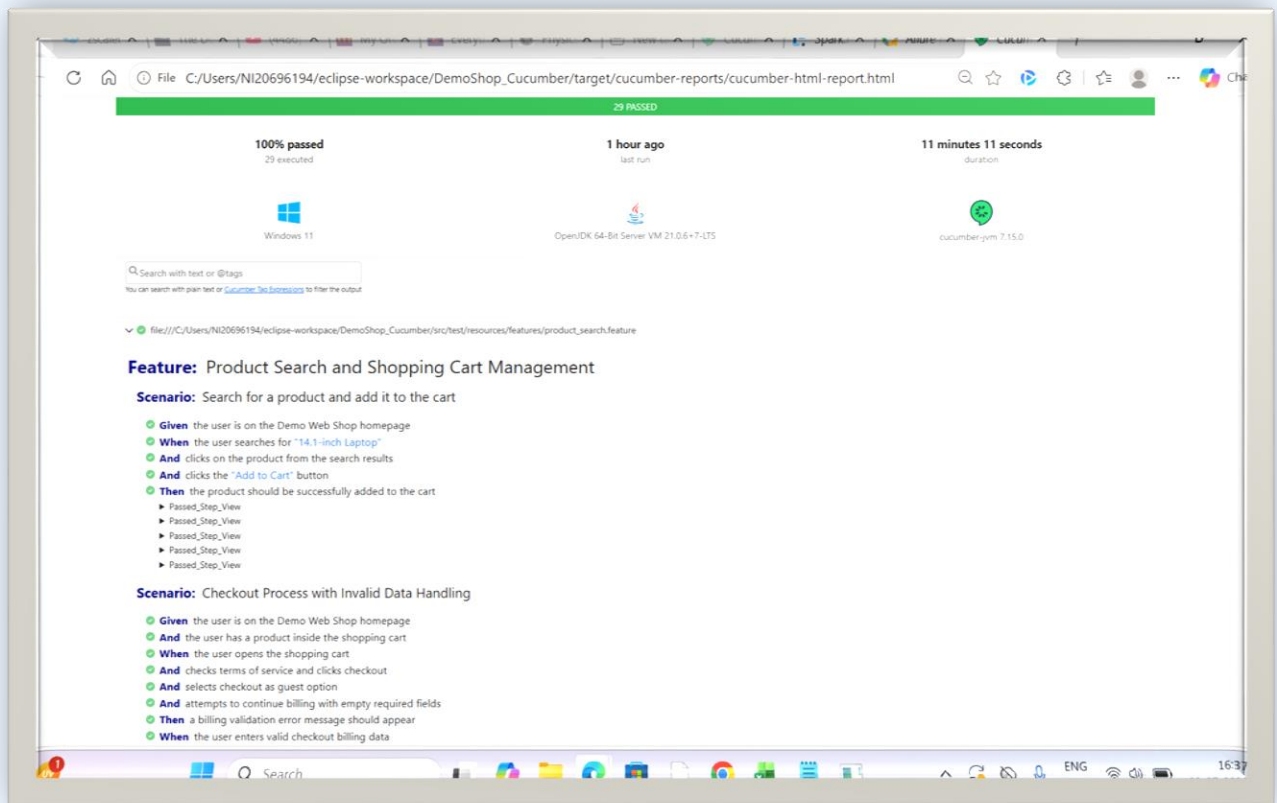


Figure 27. BDD Cucumber HTML Report

CHAPTER 11: VERSION CONTROL

GitHub was used for source code management and team collaboration.

Activities Performed

- Repository Creation
- Code Commit
- Branch Management
- Pull Requests
- Version Tracking

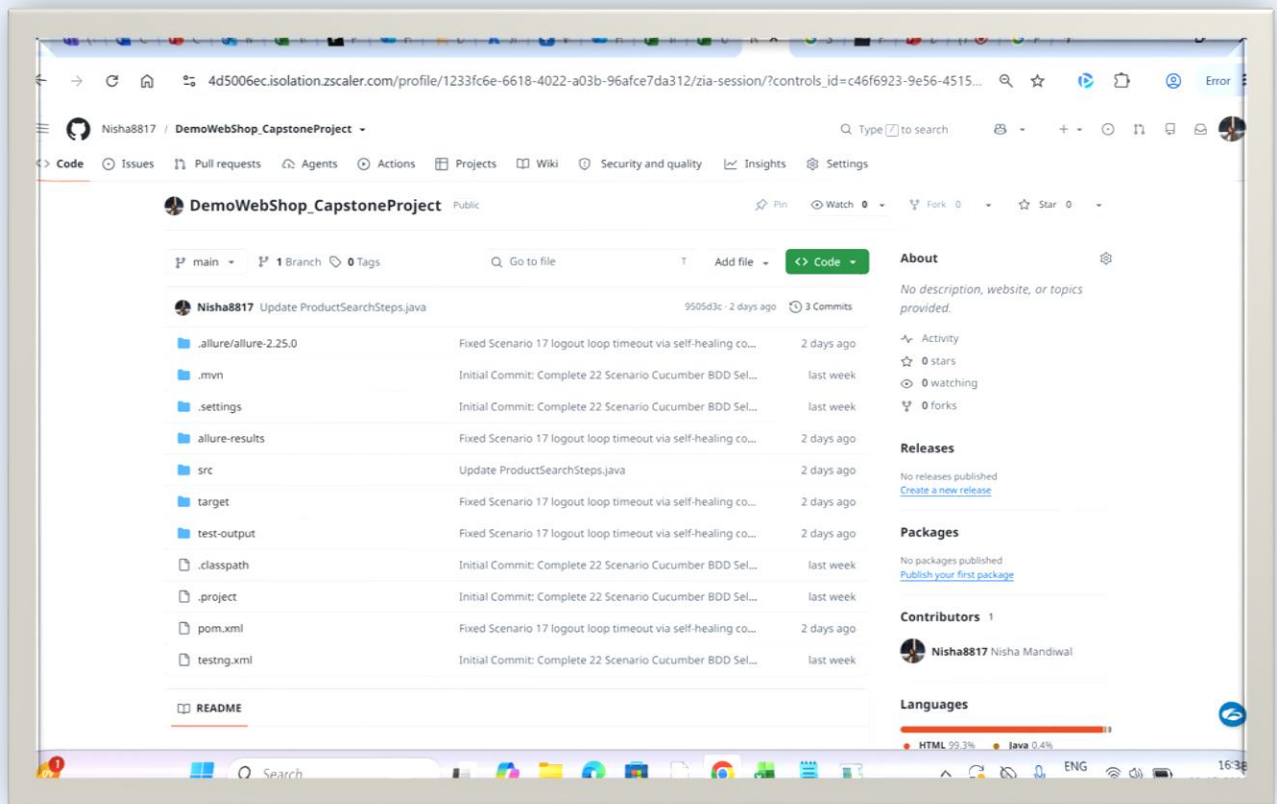


Figure 28. GitHub Repository Management

CHAPTER 12: RESULTS AND DISCUSSION

The automation framework successfully automated major functionalities of the Demo Web Shop application.

Achievements

- Improved execution speed
- Reduced manual effort
- Increased test coverage
- Enhanced maintainability
- Automated report generation
- Data-driven execution support

Challenges Addressed

- Dynamic Web Elements
- Synchronization Issues
- Test Data Management
- Framework Scalability
- Report Integration

CHAPTER 13: CONCLUSION

The Demo Web Shop Automation Framework successfully demonstrates the practical implementation of Selenium WebDriver, Cucumber BDD, TestNG, Maven, and Java in automating an e-commerce application. The framework follows Page Object Model design principles, supports data-driven testing, and generates comprehensive reports using Allure and Extent Reports.

The project provides a scalable and maintainable solution for web application testing and reflects industry-standard automation testing practices. Future enhancements may include CI/CD integration, Selenium Grid implementation, cross-browser execution, and cloud-based testing solutions.

REFERENCES

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2. Cucumber Official Documentation
3. TestNG Documentation
4. Apache POI Documentation
5. Maven Official Documentation
6. Allure Reporting Documentation
7. Extent Reports Documentation
8. Demo Web Shop Application (<https://demowebshop.tricentis.com>)

